EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. The value of ${}^{50}C_4 + \sum_{r=1}^{6} {}^{56-r}C_3$ is

(A) ${}^{56}C_4$ (B) ${}^{56}C_3$ (C) ${}^{55}C_3$ (D) ${}^{55}C_4$

MPC002

2. At an election, a voter may vote for any number of candidates, not greater than the number to be elected. There are 10 candidates and 4 are to be elected. If a voter votes for at least one candidates, then the number of ways in which he can vote is

(A) 385 (B) 1110 (C) 5040 (D) 6210

MPC003

3. The set $S = \{1, 2, 3, \ldots\ldots, 12\}$ is to partitioned into three sets A, B, C of equal size.
Thus, $A \cup B \cup C = S, A \cap B = B \cap C = C \cap A = \phi$, then number of ways to partition S are-

(A) $\frac{12!}{3!(3!)^4}$ (B) $\frac{12!}{(4!)^3}$ (C) $\frac{12!}{(3!)^4}$ (D) $\frac{12!}{3!(4!)^3}$

MPC004

4. The number of integers greater than 6000 that can be formed, using the digits 3,5,6,7 and 8 without repetition, is :

(A) 120 (B) 72 (C) 216 (D) 192

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MPC005

5. From 6 different novels and 3 different dictionaries, 4 novels and 1 dictionary are to be selected and arranged in a row on a shelf so that the dictionary is always in the middle. Then the number of such arrangements is :

(A) At least 750 but less than 1000 (B) At least 1000
(C) Less than 500 (D) At least 500 but less than 750

MPC006

6. There are two urns. Urn A has 3 distinct red balls and urn B has 9 distinct blue balls. From each urn two balls are taken out at random and then transferred to the other. The number of ways in which this can be done is:

(A) 3 (B) 36 (C) 66 (D) 108

MPC007

7. If the letters of the word $'SACHIN'$ are arranged in all possible ways and these words are written out as in dictionary, then the word $'SACHIN'$ appears at serial number

(A) 602 (B) 603 (C) 600 (D) 601

MPC001

8. The number of points, having both co-ordinates as integers, that lie in the interior of the triangle with vertices (0, 0), (0, 41) and (41, 0) is :

(A) 820 (B) 780 (C) 901 (D) 861

MPC008

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9. There are 10 points in a plane, out of these 6 are collinear. If N is the number of triangles formed by joining these points, then :
(A) $N > 190$ (B) $N \leq 100$ (C) $100 < N \leq 140$ (D) $140 < N \leq 190$

MPC009

10. Let T_n be the number of all possible triangles formed by joining vertices of an n –sided regular polygon. If $T_{n+1} - T_n = 10$, then the value of n is :
(A) 7 (B) 5 (C) 10 (D) 8

MPC012

11. Assuming the balls to be identical except for difference in colours, the number of ways in which one or more balls can be selected from 10 white, 9 green and 7 black balls is :
(A) 879 (B) 880 (C) 629 (D) 630

MPC010

12. Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of $A \times B$ having 3 or more elements is
(A) 256 (B) 220 (C) 219 (D) 211

MPC011

13. A 5 digit number divisible by 3 is to be formed using the numerals 0,1,2,3,4 & 5 without repetition. The total number of ways this can be done is –
(A) 3125 (B) 600 (C) 240 (D) 216

MPC013

14. Number of ways in which 9 different toys can be distributed among 4 children belonging to different age groups in such a way that distribution among the 3 elder children is even and the youngest one is to receive one toy more is-

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(A) $\frac{(5!)^2}{8}$ (B) $\frac{9!}{2}$ (C) $\frac{9!}{3!(2!)^3}$ (D) none of these

MPC014

15. The number of ways in which we can arrange n ladies & n gentlemen at a round table so that no 2 ladies sit together is
(A) $(n-1)!\,(n-2)!$ (B) $(n)!\,(n-1)!$
(C) $(n+1)!\,(n)!$ (D) none of these

MPC161

16. The number of ways in which 10 identical apples can be distributed among 6 children so that each child receives atleast one apple is -
(A) 126 (B) 252 (C) 378 (D) none of these

MPC016

17. The number of ways in which a mixed double tennis game can be arranged from amongst 9 married couple if no husband & wife plays in the same game is -
(A) 756 (B) 3024 (C) 1512 (D) 6048

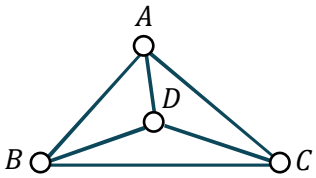
MPC017

18. Number of different words that can be formed using all the letters of the word "*DEEPMALA*" if two vowels are together and the other two are also together but separated from the first two is
(A) 960 (B) 1200 (C) 2160 (D) 1440

MPC018

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19. A road network as shown in the figure connect four cities. In how many ways can you start from any city (say A) and come back to it without travelling on the same road more than once?



(A) 8 (B) 12 (C) 9 (D) 16

MPC019

20. Sum of all the numbers that can be formed using all the digits 2, 3, 3, 4, 4, 4 is -

(A) 22222200 (B) 11111100 (C) 55555500 (D) 20333280

MPC020

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Lines $y = x + i$ & $y = -x + j$ are drawn in $x-y$ plane such that $i \in \{1,2,3,4\}$ & $j \in \{1,2,3,4,5,6\}$. If m represents the total number of squares formed by the lines and n represents the total number of triangles formed by the given lines & x-axis, then correct option/s is/are-

(A) $m + n = 50$ (B) $m - n = 2$ (C) $m + n = 48$ (D) $m - n = 4$

MPC052

2. The combinatorial coefficient $C(n, r)$ is equal to

(A) number of possible subsets of r members from a set of n distinct members.

(B) number of possible binary messages of length n with exactly r 1's.

(C) number of non decreasing $2 - D$ paths from the lattice point $(0, 0)$ to (r, n).

(D) number of ways of selecting r things out of n different things when a particular thing is always included plus the number of ways of selecting $'r'$ things out of n, when a particular thing is always excluded.

MPC053

3. There are 10 questions, each question is either True or False. Number of different sequences of incorrect answers is also equal to

(A) Number of ways in which a normal coin tossed 10 times would fall in a definite order if both Heads and Tails are present.

(B) Number of ways in which a multiple choice question containing 10 alternatives with one or more than one correct alternatives, can be answered.

(C) Number of ways in which it is possible to draw a sum of money with 10 coins of different denominations taken some or all at a time.

(D) Number of different selections of 10 indistinguishable things taken some or all at a time.

MPC054

4. Number of ways in which 3 numbers in A.P. can be selected from $1, 2, 3, \ldots\ldots n$ is :

(A) $\left(\frac{n-1}{2}\right)^2$ if n is even (B) $\frac{n(n-2)}{4}$ if n is odd

(C) $\frac{(n-1)^2}{4}$ if n is odd (D) $\frac{n(n-2)}{4}$ if n is even

MPC055

5. The combinatorial coefficient ${}^{n-1}C_p$ denotes

(A) the number of ways in which n things of which p are alike and rest different can be arranged in a circle.

(B) the number of ways in which p different things can be selected out of n different thing if a particular thing is always excluded.

(C) number of ways in which n alike balls can be distributed in p different boxes so that no box remains empty and each box can hold any number of balls.

(D) the number of ways in which $(n-2)$ white balls and p black balls can be arranged in a line if black balls are separated, balls are all alike except for the colour.

MPC056

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6. The maximum number of permutations of $2n$ letters in which there are only a's & b's, taken all at a time is given by:

(A) ${}^{2n}C_n$ (B) $\frac{2}{1}.\frac{6}{2}.\frac{10}{3}.....\frac{4n-6}{n-1}.\frac{4n-2}{n}$

(C) $\frac{n+1}{1}.\frac{n+2}{2}.\frac{n+3}{3}.\frac{n+4}{4}.....\frac{2n-1}{n-1}.\frac{2n}{n}$ (D) $\frac{2^n.\left[1.3.5.....(2n-3)(2n-1)\right]}{n!}$

MPC057

7. The number of non-negative integral solutions of $x_1 + x_2 + x_3 + x_4 \le n$ (where n is a positive integer) is

(A) ${}^{n+3}C_3$ (B) ${}^{n+4}C_4$ (C) ${}^{n+5}C_5$ (D) ${}^{n+4}C_n$

MPC162

8. The number of ways in which 200 different things can be divided into groups of 100 pairs, is :

(A) $\frac{200!}{2^{100}}$ (B) $\left(\frac{101}{2}\right)\left(\frac{102}{2}\right)\left(\frac{103}{2}\right)...\left(\frac{200}{2}\right)$

(C) $\frac{200!}{2^{100}(100)!}$ (D) (1 . 3 . 5 ... 199)

MPC163

9. The linear permutation of the word BARAAKOBAMA can be done (using all letters) -

(A) in $\frac{11!}{5!2!}$ ways if there is no constraint

(B) in ${}^6C_2 . 5!$ ways if A's are together & B's are separated

(C) in 6! 2! ways if all consonants & all vowels are together

(D) in $\frac{6!}{3!}$ ways if letter of BARAAK are arranged, keeping OBAMA fixed at extrem right position

MPC164

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COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 10 to 12

Let p be a prime number and n be a positive integer, then exponent of p in $n!$ is denoted by $E_p(n!)$ and is given by

$$E_p(n!) = \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \cdots + \left[\frac{n}{p^k}\right]$$

Where $p^k < n < p^{k+1}$

and $[x]$ denotes the integral part of x.

If we isolate the power of each prime contained in any number N, then N can be written as

$N = 2^{\alpha_1}3^{\alpha_2}5^{\alpha_3}7^{\alpha_4}...$

where α_i are whole numbers.

On the basis of above information, answer the following questions:

10. The exponent of 7 in ${}^{100}C_{50}$ is -

(A) 0 (B) 1 (C) 2 (D) 3

MPC061

For More Join: @IITJEE_Advanced

11. The number of zeros at the end of 108! is -

(A) 10 (B) 13 (C) 25 (D) 26

MPC062

12. The exponent of 12 in 100! is -

(A) 32 (B) 48 (C) 97 (D) none of these

MPC063

MATRIX MATCH TYPE QUESTION

Answer Q.13, Q.14 and Q.15 by appropriately matching the information given in the three columns of the following table

Column 1, 2 & 3 consists of different words, number of selection of 3 letters from the word and number of arrangements of 3 letters from the word respectively.

Column-1	**Column-2**	**Column-3**
(I) DREAM	(i) 70	(P) 60
(II) DEDICATION	(ii) 10	(Q) 399
(III) POWERFUL	(iii) 77	(R) 336
(IV) COMBINATION	(iv) 56	(S) 378

13. Which of the following is a correct combination?

(A) (I) (i) (P) (B) (III) (iv) (R) (C) (II) (ii) (S) (D) (IV) (ii) (Q)

MPC058

14. Which of the following is correct combination?

(A) (III) (iv) (S) (B) (III) (i) (R) (C) (IV) (iii) (Q) (D) (I) (ii) (R)

MPC059

15. Which of the following is correct combination?

(A) (I) (ii) (S) (B) (II) (i) (Q) (C) (III) (iv) (R) (D) (IV) (iii) (R)

MPC060

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EXERCISE (S)

1. An examination paper consists of 12 questions divided into parts A & B. Part $-A$ contains 7 questions & Part-B contains 5 questions. If a candidate is required to attempt 8 questions selecting atleast 3 from each part, then number of maximum ways can the candidate select the questions are

MPC024

2. Let 5 boys & 4 girls sit in a straight line. If the number of ways in which they can be seated if 2 girls are together & the other 2 are also together but separated from the first 2 are N, then sum of digits of N is

MPC025

3. There are 2 women participating in a chess tournament. Every participant played 2 games with the other participants. The number of games that the men played between themselves exceeded by 66 as compared to the number of games that the men played with the women. If number of participants & the total number of games played in the tournament are M and N respectively, then value of $(M+N)$ is

MPC026

4. Find the ratio ${}^{20}C_r$ to ${}^{25}C_p$ when each of them has the greatest value possible.

MPC165

5. Let P_n denotes the number of ways in which three people can be selected out of 'n' people sitting in a row, if no two of them are consecutive. If $P_{n+1} - P_n = 15$ then the value of 'n' is

MPC021

6. The number of positive integral solutions of the equation $x_1x_2x_3 = 60$ is

MPC022

7. Total number of even divisors of 2079000 which are divisible by 15 are

MPC023

8. If the sum of all numbers greater than 10000 formed by using the digits 0, 1, 2, 4, 5 & no digit being repeated in any number, is N, then sum of digits of N is

MPC027

9. Let the number of ways in which the letters of the word 'MUNMUN' can be arranged so that no two alike letters are together are N, then value of N is

MPC028

10. If number of integral solutions for the equation; $x+y+z+w=29$ when $x>0, y>1, z>2$ & $w\geq 0$ is equal to N, then find the value of $\frac{N}{100}$?

MPC030

11. Six persons A, B, C, D, E and F are to be seated at a circular table. The number of ways this can be done if A must have either B or C on his right and B must have either C or D on his right is -

MPC049

12. 5 Indian & 5 American couples meet at a party & shake hands. If no wife shakes hands with her own husband & no Indian wife shakes hands with a male, then the number of hand shakes that takes place in the party is –

MPC050

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For More Join: @IITJEE_Advanced

13. The number of ways of choosing a committee of 2 women & 3 men from 5 women & 6 men, if Mr. A refuses to serve on the committee if Mr. B is a member & Mr. B can only serve, if Miss C is the member of the committee, is -

MPC051

14. The number of permutations which can be formed out of the letters of the word "SERIES" taking three letters together is

MPC138

15. A box contains 6 balls which may be all of different colours or three each of two colours or two each of three different colours. The number of ways of selecting 3 balls from the box (if ball of same colour are identical) is

MPC139

16. The number of ways selecting 8 books from a library which has 10 books each of Mathematics, Physics, Chemistry and English, if books of the same subject are alike, is $(N^2 - 4)$ then N is equal to

MPC140

17. Let P_n denotes the number of ways in which three people can be selected out of 'n' people sitting in a row, if no two of them are consecutive. If, $P_{n+1} - P_n$ = 15 then the value of ${}^nC_2$ is :

MPC160

18. Number of cyphers at the end of ${}^{2002}C_{1001}$ is

MPC141

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. The number of four-digit numbers strictly greater than 4321 that can be formed using the digits 0,1,2,3,4,5 (repetition of digits is allowed) is : **[JEE (Main) 2019]**

(1) 288 (2) 306 (3) 360 (4) 310

MPC041

2. A committee of 11 members is to be formed from 8 males and 5 females. If m is the number of ways the committee is formed with at least 6 males and n is the number of ways the committee is formed with at least 3 females, then : **[JEE (Main) 2019]**

(1) $m = n = 78$ (2) $n = m - 8$ (3) $m + n = 68$ (4) $m = n = 68$

MPC042

3. Some identical balls are arranged in rows to form an equilateral triangle. The first row consists of one ball, the second row consists of two balls and so on. If 99 more identical balls are added to the total number of balls used in forming the equilateral triangle, then all these balls can be arranged in a square whose each side contains exactly 2 balls less than the number of balls each side of the triangle contains. Then the number of balls used to form the equilateral triangle is:- **[JEE (Main) 2019]**

(1) 190 (2) 262 (3) 225 (4) 157

MPC091

4. The number of natural numbers less than 7,000 which can be formed by using the digits 0,1,3,7,9 (repetition of digits allowed) is equal to : **[JEE (Main) 2019]**

(1) 250 (2) 374 (3) 372 (4) 375

MPC092

Rankers Academy JEE

5. Let $n > 2$ be an integer. Suppose that there are n Metro stations in a city located along a circular path. Each pair of stations is connected by a straight track only. Further, each pair of nearest stations is connected by blue line, whereas all remaining pairs of stations are connected by red line. If the number of red lines is 99 times the number of blue lines, then the value of n is : **[JEE (Main) 2020]**

(1) 199 (2) 101 (3) 201 (4) 200

MPC043

6. There are 3 sections in a question paper and each section contains 5 questions. A candidate has to answer a total of 5 questions, choosing at least one question from each section. Then the number of ways, in which the candidate can choose the questions, is : **[JEE (Main) 2020]**

(1) 1500 (2) 2255 (3) 3000 (4) 2250

MPC044

7. If the number of five digit numbers with distinct digits and 2 at the 10th place is 336 k, then k is equal to : **[JEE (Main) 2020]**

(1) 8 (2) 6 (3) 4 (4) 7

MPC093

8. If the letters of the word 'MOTHER' be permuted and all the words so formed (with or without meaning) be listed as in a dictionary, then the position of the word 'MOTHER' is _____. **[JEE (Main) 2020]**

MPC094

For More Join: @IITJEE_Advanced

9. If the digits are not allowed to repeat in any number formed by using the digits 0, 2, 4, 6, 8, then the number of all numbers greater than 10,000 is equal to _____. **[JEE (Main) 2021]**

MPC095

10. There are 5 students in class 10, 6 students in class 11 and 8 students in class 12. If the number of ways, in which 10 students can be selected from them so as to include at least 2 students from each class and at most 5 students from the total 11 students of class 10 and 11 is 100 k, then k is equal to. **[JEE (Main) 2021]**

MPC096

11. The sum of all 3-digit numbers less than or equal to 500, that are formed without using the digit "1" and they all are multiple of 11, is _____. **[JEE (Main) 2021]**

MPC097

12. The sum of all the 4-digit distinct numbers that can be formed with the digits 1, 2, 2 and 3 is: **[JEE (Main) 2021]**

(1) 26664 (2) 122664 (3) 122234 (4) 22264

MPC098

13. The letters of the word 'MANKIND' are written in all possible orders and arranged in serial order as in an English dictionary. Then the serial number of the word 'MANKIND' is _____. **[JEE (Main) 2022]**

MPC099

14. There are ten boys $B_1, B_2, ..., B_{10}$ and five girls $G_1, G_2, ..., G_5$ in a class. Then the number of ways of forming a group consisting of three boys and three girls, if both B_1 and B_2 together should not be the members of a group, is_____. **[JEE (Main) 2022]**

MPC100

15. Let S be the set of all passwords which are six to eight characters long, where each character is either an alphabet from $\{A, B, C, D, E\}$ or a number from $\{1, 2, 3, 4, 5\}$ with the repetition of characters allowed. If the number of passwords in S whose at least one character is a number from $\{1, 2, 3, 4, 5\}$ is $\alpha \times 5^6$, then α is equal to ____. **[JEE (Main) 2022]**

MPC101

16. The total number of four digit numbers such that each of the first three digits is divisible by the last digit, is equal to_____. **[JEE (Main) 2022]**

MPC102

17. The number of ways , in which 5 girls and 7 boys can be seated at a round table so that no two girls sit together, is **[JEE (Main) 2023]**

(1) $126(5!)^2$ (2) $7(360)^2$ (3) 720 (4) $7(720)^2$

MPC103

18. A person forgets his 4-digit ATM pin code. But he remembers that in the code all the digits are different, the greatest digit is 7 and the sum of the first two digits is equal to the sum of the last two digits. Then the maximum number of trials necessary to obtain the correct code is___ **[JEE (Main) 2023]**

MPC104

19. The number of numbers, strictly between 5000 and 10000 can be formed using the digits 1,3,5,7,9 without repetition, is **[JEE (Main) 2023]**

(1) 6 (2) 12 (3) 120 (4) 72

MPC105

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20. If the number of words, with or without meaning, which can be made using all the letters of the word MATHEMATICS in which C and S do not come together, is $(6!)k$, then k is equal to
[JEE (Main) 2023]
(1) 1890 (2) 945 (3) 2835 (4) 5670
MPC106

21. There are 4 men and 5 women in Group A, and 5 men and 4 women in Group B. If 4 persons are selected from each group, then the number of ways of selecting 4 men and 4 women is .....
[JEE (Main) 2024]
MPC166

22. 60 words can be made using all the letters of the word $BHBJO$, with or without meaning. If these words are written as in a dictionary, then the 50th word is : **[JEE (Main) 2024]**
(1) $OBBHJ$ (2) $HBBJO$ (3) $OBBJH$ (4) $JBBOH$
MPC167

23. Let the set $S = \{2, 4, 8, 16, ..., 512\}$ be partitioned into 3 sets A, B, C with equal number of elements such that $A \cup B \cup C = S$ and $A \cap B = B \cap C = A \cap C = \phi$. The maximum number of such possible partitions of S is equal to : **[JEE (Main) 2024]**
(1) 1680 (2) 1520 (3) 1710 (4) 1640
MPC168

24. The number of ways of getting a sum 16 on throwing a dice four times is. **[JEE (Main) 2024]**
MPC169

25. The lines $L_1, L_2, ..., I_{20}$ are distinct. For $n = 1, 2, 3, ..., 10$ all the lines L_{2n-1} are parallel to each other and all the lines L_{2n} pass through a given point P. The maximum number of points of intersection of pairs of lines from the set $\{L_1, L_2, ..., L_{20}\}$ is equal to : **[JEE (Main) 2024]**
MPC170

26. The number of elements in the set
$S = \{(x, y, z) : x, y, z \in \boldsymbol{Z}, x + 2y + 3z = 42, x, y, z \geq 0\}$ equals _____. **[JEE (Main) 2024]**
MPC171

27. Number of ways of arranging 8 identical books into 4 identical shelves where any number of shelves may remain empty is equal to **[JEE (Main) 2024]**
(1) 18 (2) 16 (3) 12 (4) 15
MPC172

28. In an examination of Mathematics paper, there are 20 questions of equal marks and the question paper is divided into three sections : A, B and C . A student is required to attempt total 15 questions taking at least 4 questions from each section. If section A has 8 questions, section B has 6 questions and section C has 6 questions, then the total number of ways a student can select 15 questions is.
[JEE (Main) 2024]
MPC173

29. The total number of words (with or without meaning) that can be formed out of the letters of the word 'DISTRIBUTION' taken four at a time, is equal to ____ **[JEE (Main) 2024]**
MPC174

30. The number of integers, between 100 and 1000 having the sum of their digits equals to 14, is _____.
[JEE (Main) 2024]
MPC175

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EXERCISE - JEE (Advanced) PYQ

1. A pack contains n cards numbered from 1 to n. Two consecutive numbered cards are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller to the numbers on the removed cards is k, then $k - 20 =$

[JEE (Advanced) 2013]

MPC107

2. Let $n_1 < n_2 < n_3 < n_4 < n_5$ be positive integers such that $n_1 + n_2 + n_3 + n_4 + n_5 = 20$. The number of such distinct arrangements $(n_1, n_2, n_3, n_4, n_5)$ is **[JEE (Advanced) 2014]**

MPC108

3. Let $n \geq 2$ b an integer. Take n distinct points on a circle and join each pair of points by a line segment. Colour the line segment joining every pair of adjacent points by blue and the rest by red. If the number of red and blue line segments are equal, then the value of n is **[JEE (Advanced) 2014]**

MPC109

4. Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and moreover the card numbered 1 is always placed in envelope numbered 2. Then the number of ways it can be done is - **[JEE (Advanced) 2014]**

(A) 264 (B) 265 (C) 53 (D) 67

MPC110

5. Let n be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that all the girls stand consecutively in the queue. Let m be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that exactly four girls stand consecutively in the queue. Then the value of $\frac{m}{n}$ is **[JEE (Advanced) 2015]**

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MPC111

6. A debate club consists of 6 girls and 4 boys. A team of 4 members is to be selected from this club including the selection of a captain (from among these 4 member) for the team. If the team has to include at most one boy, then the number of ways of selecting the team is

[JEE (Advanced) 2016]

(A) 380 (B) 320 (C) 260 (D) 95

MPC064

7. Words of length 10 are formed using the letters $A, B, C, D, E, F, G, H, I, J$. Let x be the number of such words where no letter is repeated; and let y be the number of such words where exactly one letter is repeated twice and no other letter is repeated. Then, $\frac{y}{9x} =$ **[JEE (Advanced) 2017]**

MPC065

8. Let $S = \{1, 2, 3, ..., 9\}$. For $k = 1, 2, ..., 5$, let N_k be the number of subsets of S, each containing five elements out of which exactly k are odd. Then $N_1 + N_2 + N_3 + N_4 + N_5 =$

[JEE (Advanced) 2017]

(A) 125 (B) 252 (C) 210 (D) 126

MPC066

For More Join: @IITJEE_Advanced

9. The number of 5 digit numbers which are divisible by 4, with digits from the set {1, 2, 3, 4, 5} and the repetition of digits is allowed, is. **[JEE (Advanced) 2018]**

MPC067

10. Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one-one functions from X to Y and β is the number of onto functions from Y to X, then the value of $\frac{1}{5!}(\beta - \alpha)$ is ____. **[JEE (Advanced) 2018]**

MPC068

11. In a high school, a committee has to be formed from a group of 6 boys $M_1, M_2, M_3, M_4, M_5, M_6$ and 5 girls G_1, G_2, G_3, G_4, G_5.

(i) Let α_1 be the total number of ways in which the committee can be formed such that the committee has 5 members, having exactly 3 boys and 2 girls.

(ii) Let α_2 be the total number of ways in which the committee can be formed such that the committee has at least 2 members, and having an equal number of boys and girls.

(iii) Let α_3 be the total number of ways in which the committee can be formed such that the committee has 5 members, at least 2 of them being girls.

(iv) Let α_4 be the total number of ways in which the committee can be formed such that the committee has 4 members, having at least 2 girls and such that both M_1 and G_1 are **NOT** in the committee together.

LIST-I	**LIST-II**
P. The value of α_1 is	1. 136
Q. The value of α_2 is	2. 189
R. The value of α_3 is	3. 192
S. The value of α_4 is	4. 200
	5. 381
	6. 461

[JEE (Advanced) 2018]

The correct option is :

(A) P $\rightarrow$ 4; Q $\rightarrow$ 6, R $\rightarrow$ 2; S $\rightarrow$ 1

(B) P $\rightarrow$ 1; Q $\rightarrow$ 4; R $\rightarrow$ 2; S $\rightarrow$ 3

(C) P $\rightarrow$ 4; Q $\rightarrow$ 6, R $\rightarrow$ 5; S $\rightarrow$ 2

(D) P $\rightarrow$ 4; Q $\rightarrow$ 2; R $\rightarrow$ 3; S $\rightarrow$ 1

MPC069

12. Five person A, B, C, D and E are seated in a circular arrangement. If each of them is given a hat of one of the three colours red, blue and green, then the number of ways of distributing the hats such that the persons seated in adjacent seats get different coloured hats is **[JEE (Advanced) 2019]**

MPC070

13. An engineer is required to visit a factory for exactly four days during the first 15 days of every month and it is mandatory that **no** two visits take place on consecutive days. Then the number of all possible ways in which such visits to the factory can be made by the engineer during 1–15 June 2021 is ____. **[JEE (Advanced) 2020]**

MPC071

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14. In a hotel, four rooms are available. Six persons are to be accommodated in these four rooms in such a way that each of these rooms contains at least one person and at most two persons. Then the number of all possible ways in which this can be done is **[JEE (Advanced) 2020]**

MPC072

15. Let

$S_1 = \{(i, j, k) : i, j, k \in \{1, 2,...,10\}\}$

$S_2 = \{(i, j) : 1 \le i < j + 2 \le 10,\ i, j \in \{1, 2,...,10\}\}$,

$S_3 = \{(i, j, k, l) : 1 \le i < j < k < l,\ i, j, k, l \in \{1, 2,....,10\}\}$.

and

$S_4 = \{(i, j, k, l) : i, j, k$ and l are distinct elements in $\{1, 2,...,10\}\}$.

If the total number of elements in the set S_r is n_r, $r = 1, 2, 3, 4$, then which of the following statements is (are) **TRUE**? **[JEE (Advanced) 2021]**

(A) $n_1 = 1000$ (B) $n_2 = 44$ (C) $n_3 = 220$ (D) $\frac{n_4}{12} = 420$

MPC088

16. The number of 4-digit integers in the closed interval [2022, 4482] formed by using the digits 0, 2, 3, 4, 6, 7 is. **[JEE (Advanced) 2022]**

MPC089

17. Consider 4 boxes, where each box contains 3 red balls and 2 blue balls. Assume that all 20 balls are distinct. In how many different ways can 10 balls be chosen from these 4 boxes so that from each box at least one red ball and one blue ball are chosen? **[JEE (Advanced) 2022]**

(A) 21816 (B) 85536 (C) 12096 (D) 156816

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MPC090

18. A group of 9 students, $s_1, s_2,....., s_9$, is to be divided to from three teams X, Y, and Z of sizes 2, 3, and 4, respectively. Suppose that s_1 cannot be selected for the team X, and s_2 cannot be selected for the team Y. Then the number of ways to from such teams, is _____. **[JEE (Advanced) 2024]**

MPC176

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	B	D	B	D	D	B	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	C	D	C	B	A	C	D	B	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	A,B,D	B,C	C,D	B,D	A,B,C,D	B,D	B,C,D	A,B,D	A
Que.	11	12	13	14	15					
Ans.	C	B	B	C	C					

EXERCISE (S)

1.	420	**2.**	9	**3.**	169	**4.**	420	**5.**	8
6.	54	**7.**	108	**8.**	36	**9.**	30	**10.**	26
11.	18	**12.**	135	**13.**	124	**14.**	42	**15.**	31
16.	13	**17.**	28	**18.**	1				

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	1	2	3	4	1	309	96	238
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	7744	1	1492	1120	7073	1086	1	72	4	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	5626	3	1	125	101	169	4	11376	3734	70

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	5	7	5	C	5	A	5	D	625	119
Que.	11	12	13	14	15	16	17	18		
Ans.	C	30	495	1080	A,B,D	569	A	665		

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